Stochastic Inventory Optimization using Markov Decision Processes

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Abstract

This report studies a stochastic inventory control problem using a Markov Decision Process (MDP) framework. The goal is to determine optimal ordering decisions under uncertain demand while balancing holding costs, shortage penalties, and ordering costs.

Demand is modeled as a stochastic process and the inventory system evolves dynamically across time periods. Value iteration is used to compute optimal ordering policies. Monte Carlo simulations are performed to evaluate system performance and compare optimized policies against naive decision rules.

1 Introduction

Inventory management is a fundamental problem in supply chain systems. Firms must determine how much inventory to order while facing uncertain demand patterns.

Ordering excessive inventory leads to large holding costs while insufficient inventory leads to shortages and service failures.

Stochastic optimization techniques allow decision-makers to account for demand uncertainty. In this work, the inventory system is modeled as a Markov Decision Process where the state corresponds to the inventory level and actions correspond to ordering decisions.

The objective is to minimize long-run expected operational cost.

2 Stochastic Demand Model

Demand is modeled as a Poisson random variable

$$D_t \sim Poisson(\lambda)$$

where λ represents the expected demand rate.

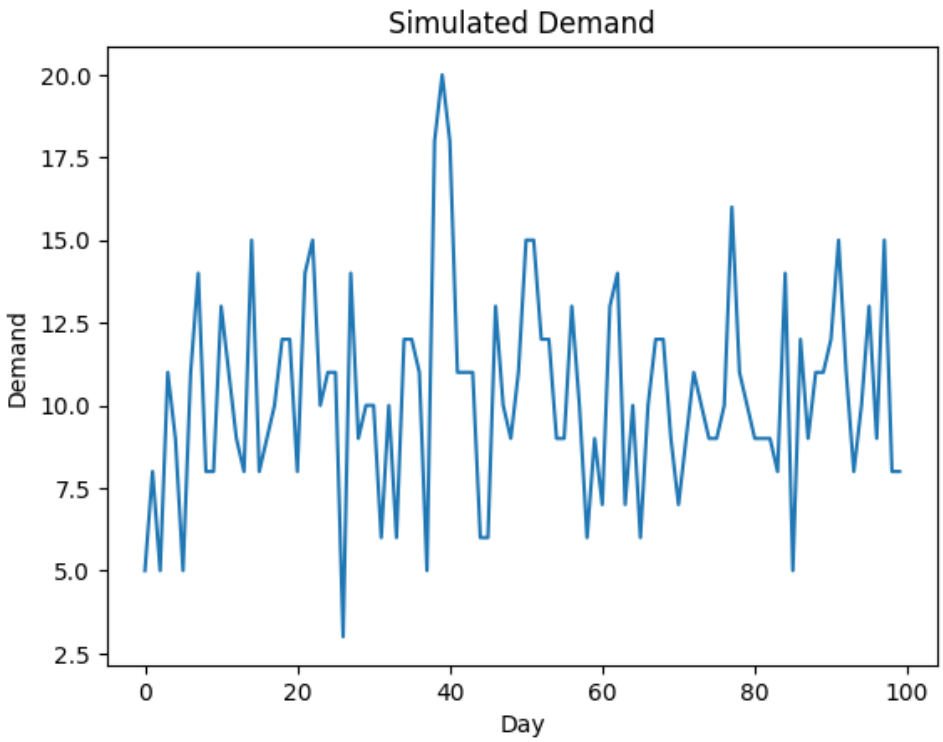

Figure 1: Simulated stochastic demand

3 Inventory Dynamics

Let s_t denote the inventory level at time t. The system evolves as

$$s_{t+1} = \max(s_t + a_t - D_t, 0)$$

where

- a_t = order quantity
- D_t = stochastic demand

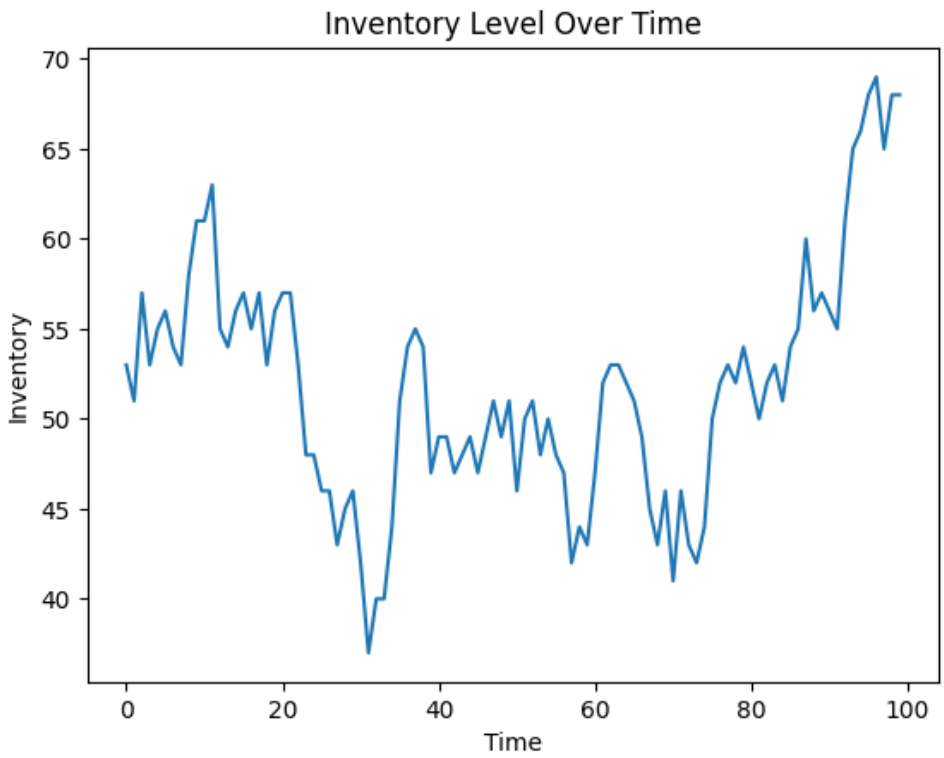

Figure 2: Inventory trajectory

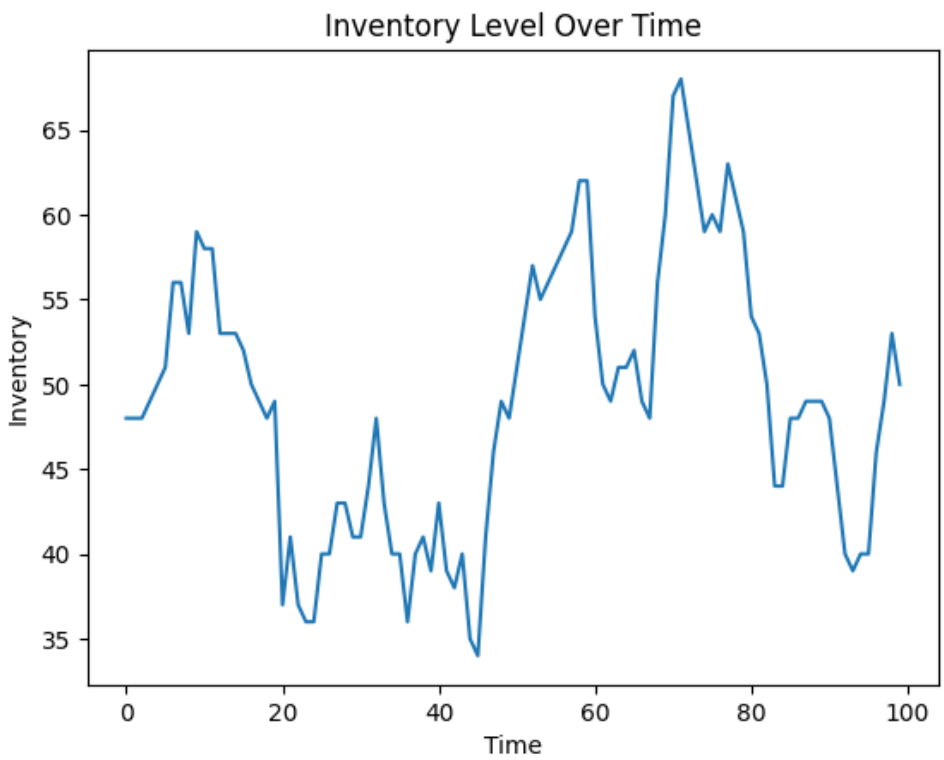

Figure 3: Inventory dynamics under stochastic demand

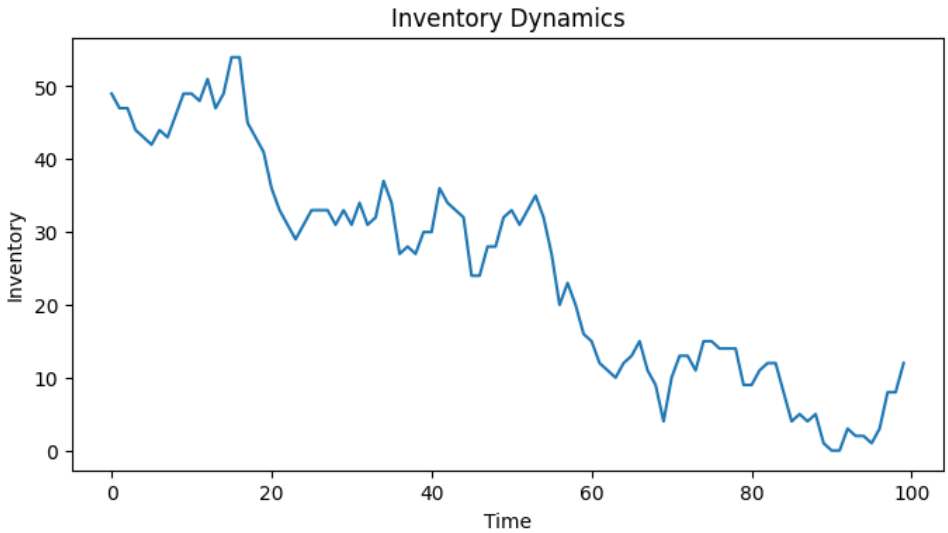

Figure 4: Inventory fluctuations

4 Cost Function

The system incurs three types of costs

- Holding cost
- Shortage penalty
- Ordering cost

The cost function is

$$C(s, a, D) = h \max(s - D, 0) + p \max(D - s, 0) + ca$$

where

- h = holding cost
- p = shortage penalty
- c = ordering cost

5 Markov Decision Process

The inventory system is modeled as a Markov Decision Process defined by

- State space S (inventory levels)
- Action space A (order quantities)
- Transition probabilities
- Cost function

The value function satisfies the Bellman equation

$$V(s) = \min_a E[C(s, a, D) + \gamma V(s')]$$

Value iteration is used to compute optimal policies.

6 Optimal Ordering Policy

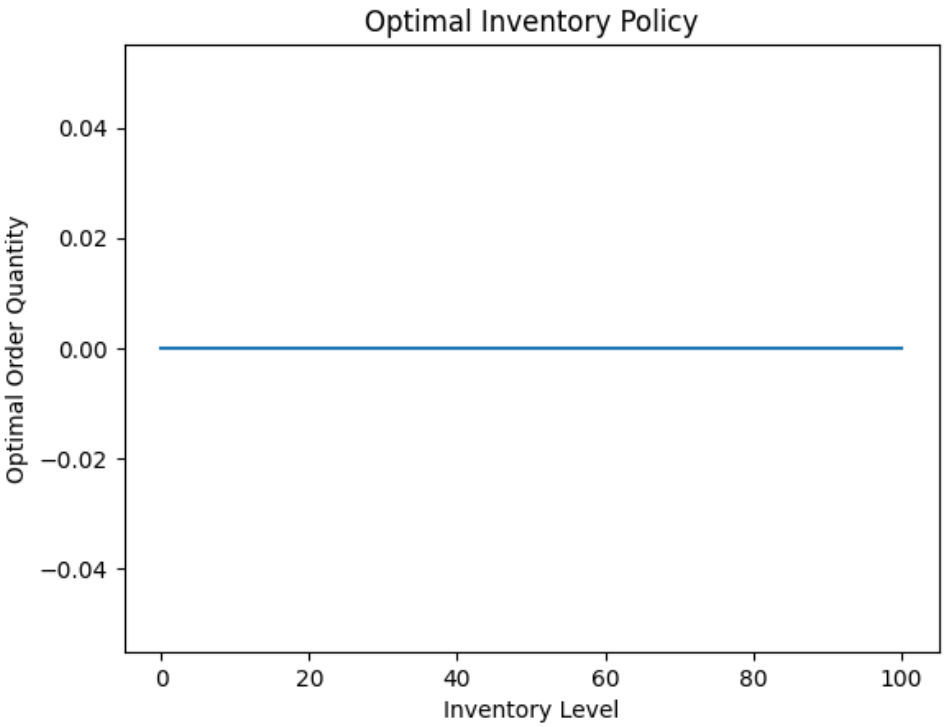

Figure 5: Initial ordering policy

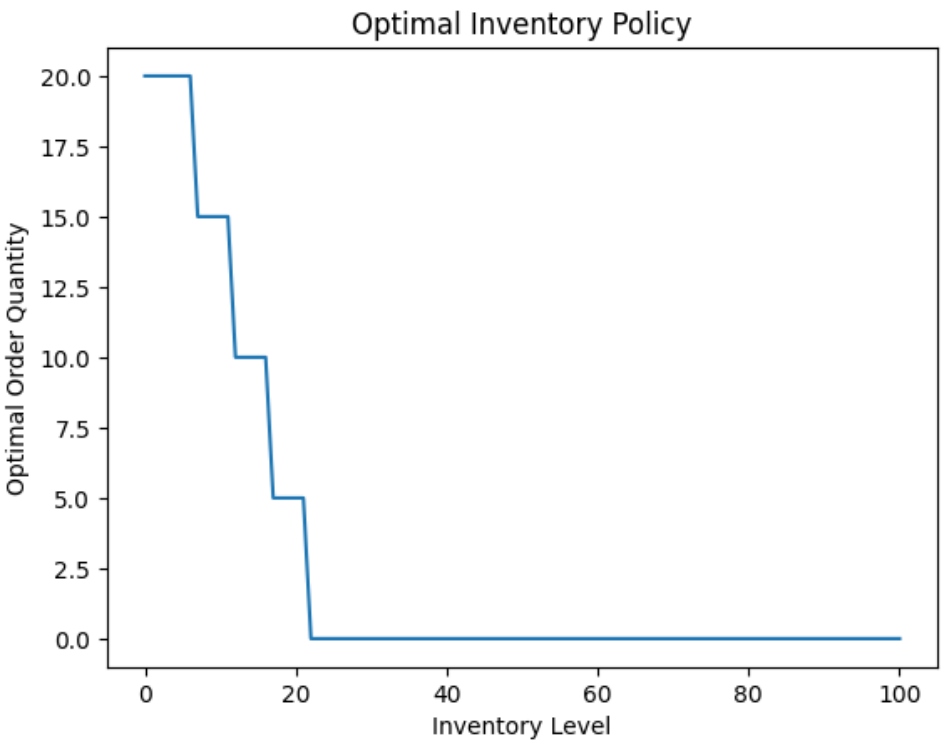

Figure 6: Optimal ordering policy

7 Two Stage Stochastic Optimization

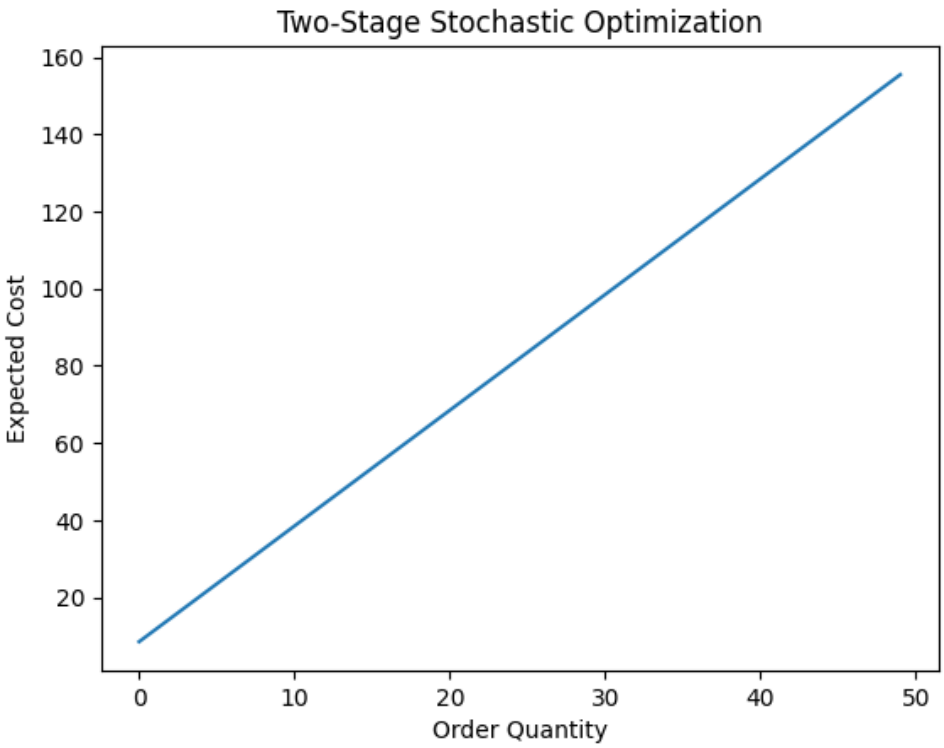

Figure 7: Expected cost vs order quantity

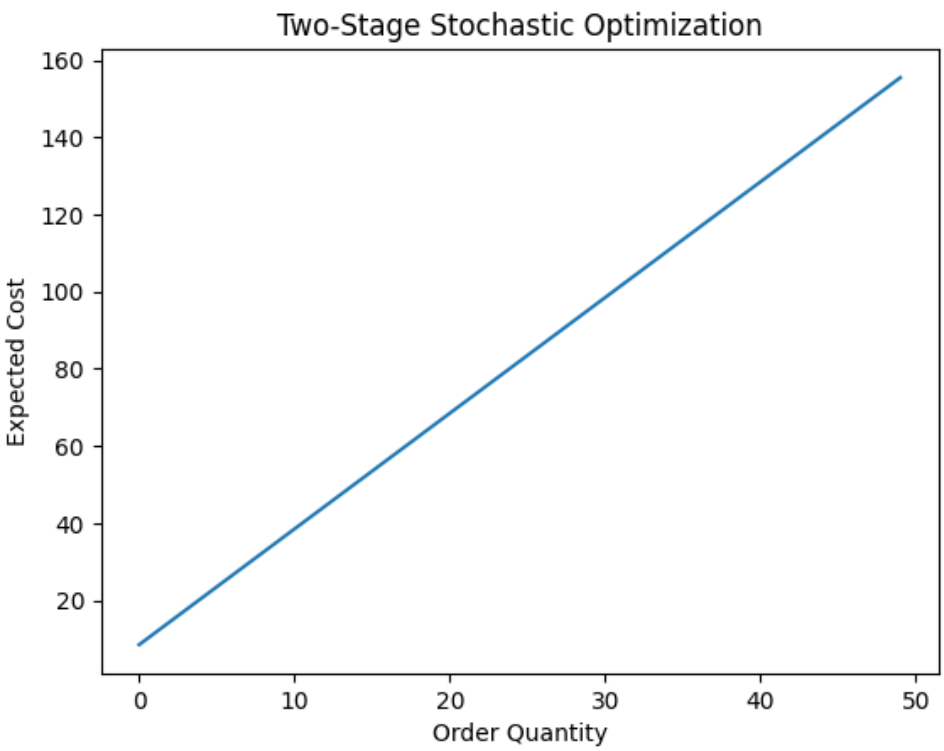

Figure 8: Two-stage stochastic optimization

8 Policy Comparison

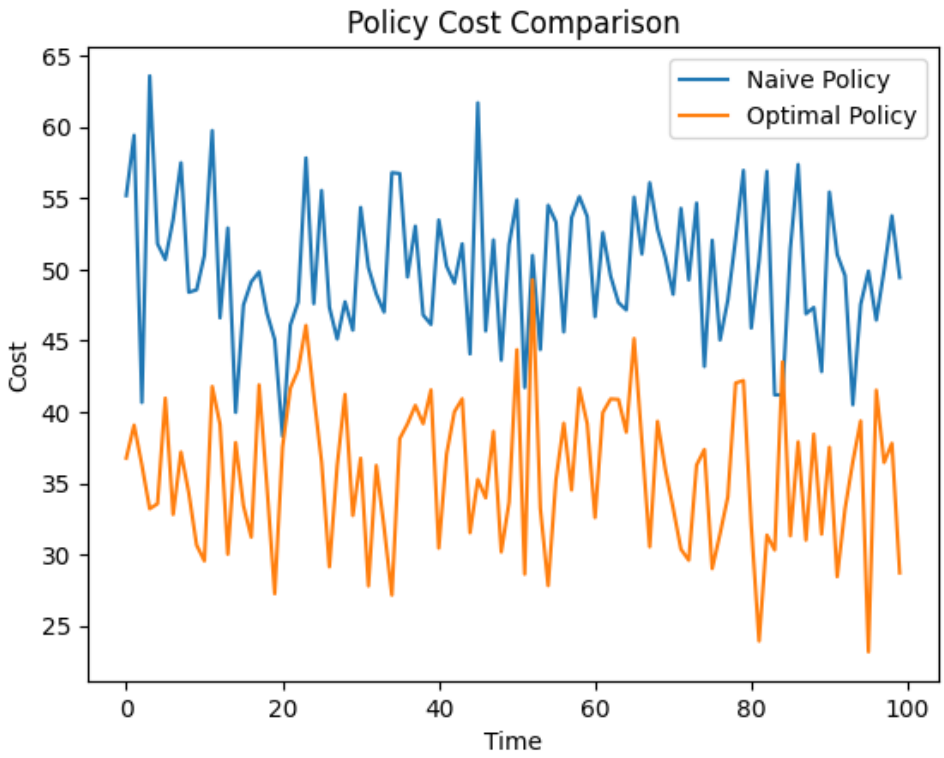

Figure 9: Cost comparison between policies

9 Monte Carlo Simulation

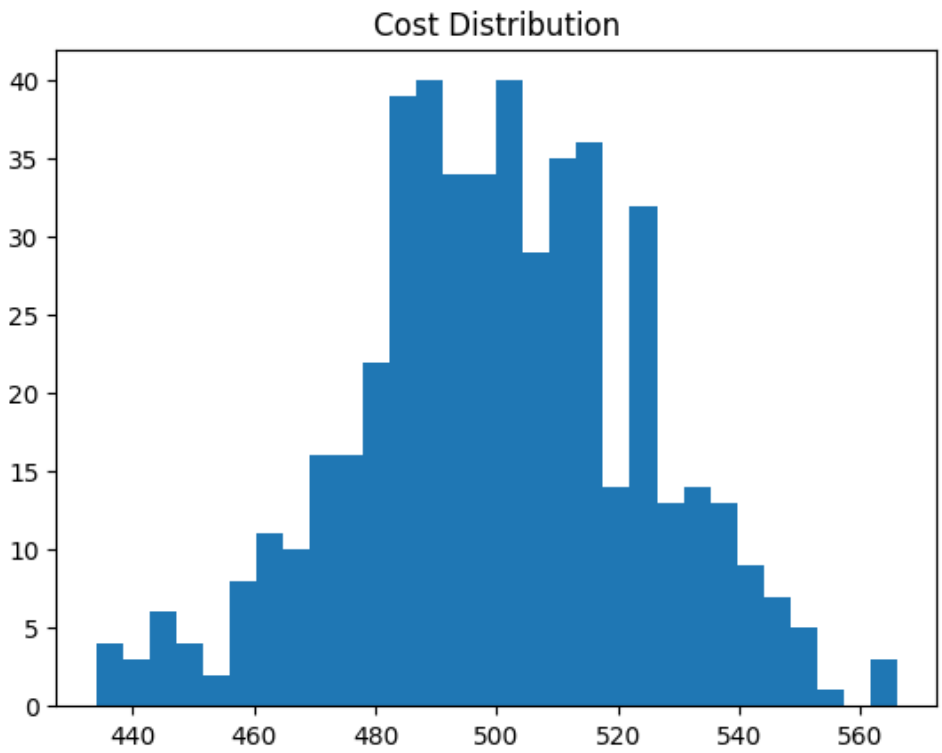

Figure 10: Monte Carlo cost distribution

10 Policy Visualization

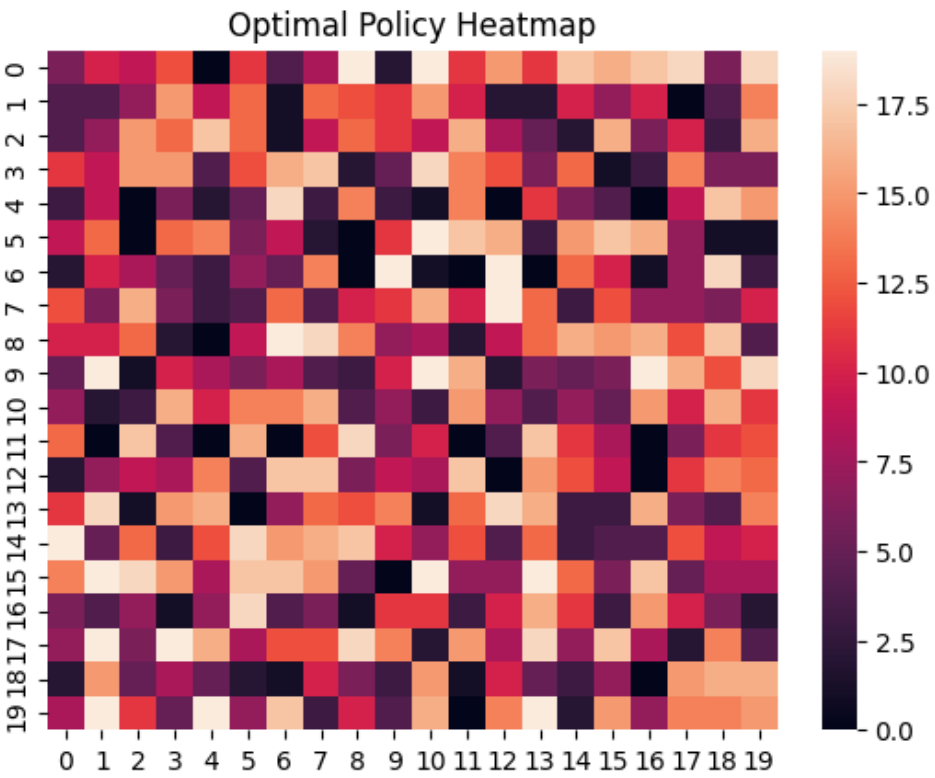

Figure 11: Policy heatmap

11 Sensitivity Analysis

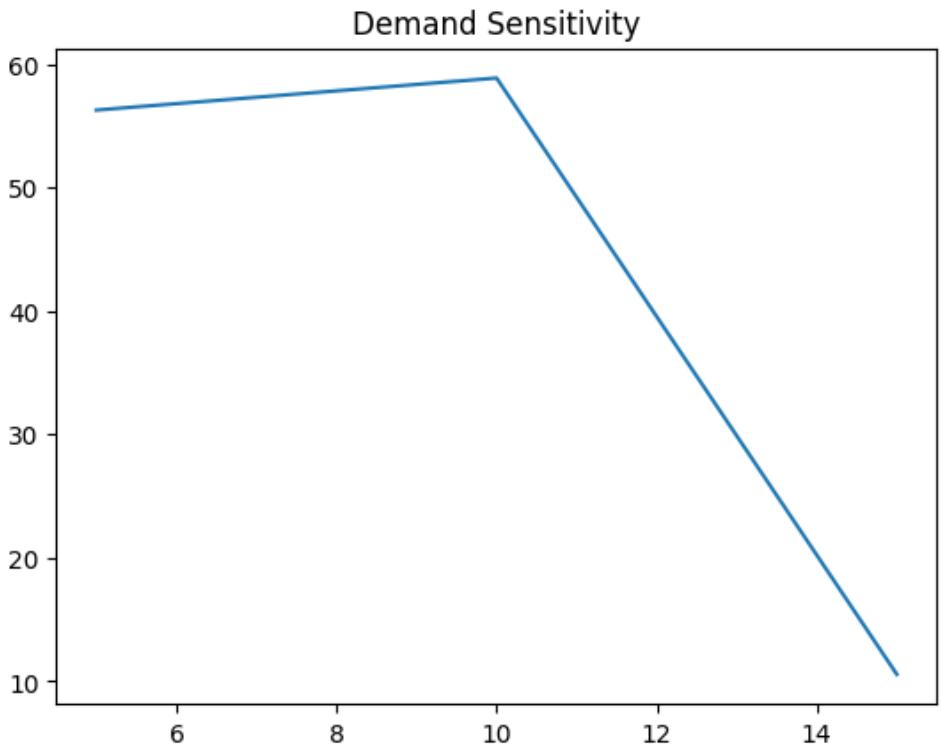

Figure 12: Demand sensitivity

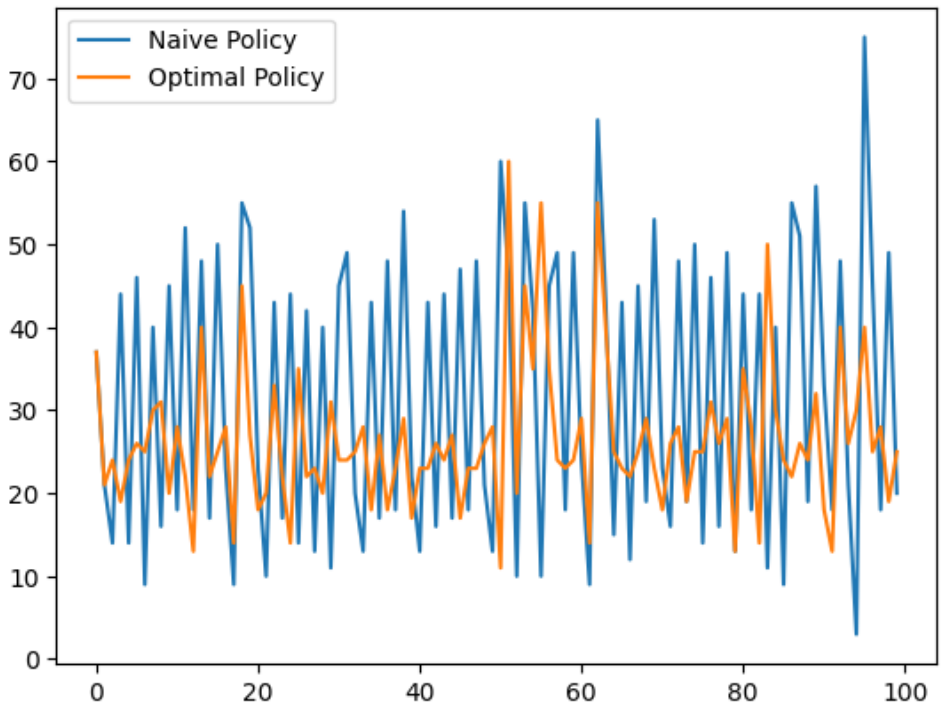

Figure 13: Policy time series

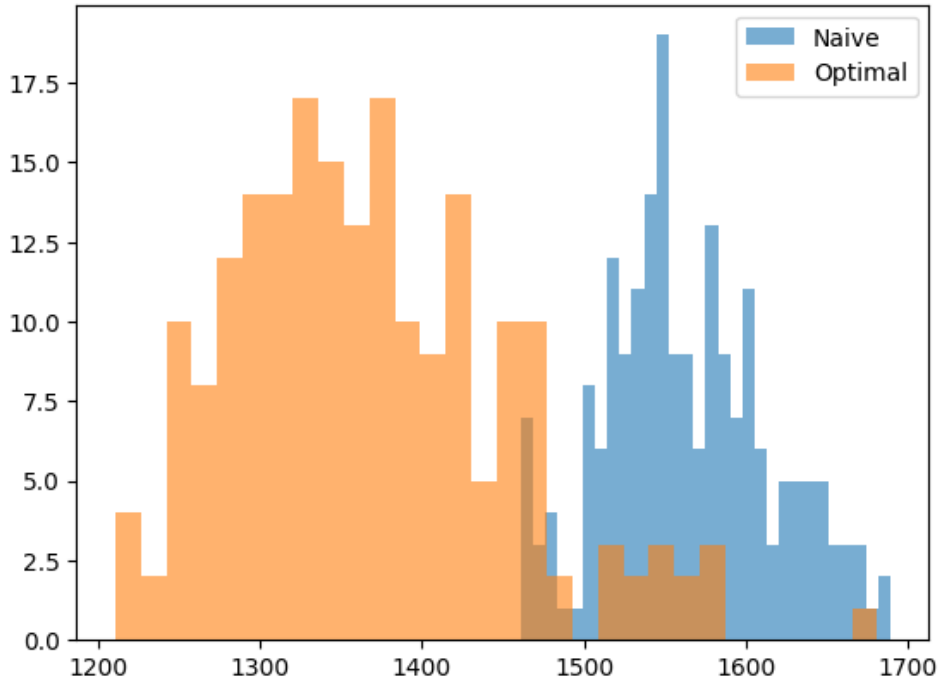

Figure 14: Cost distribution comparison

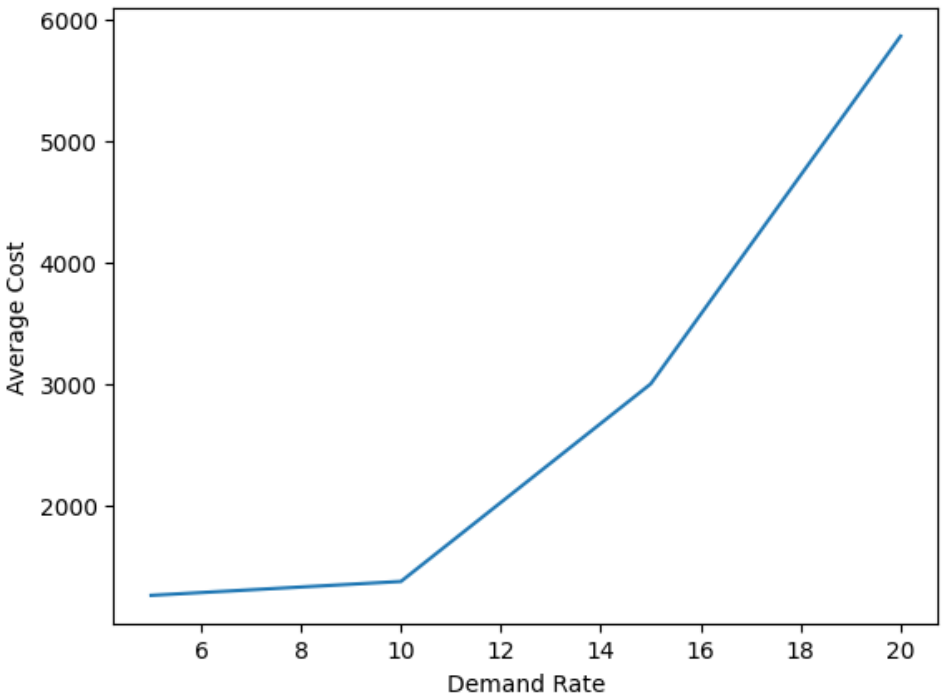

Figure 15: Average cost vs demand rate

12 Inventory Time Series

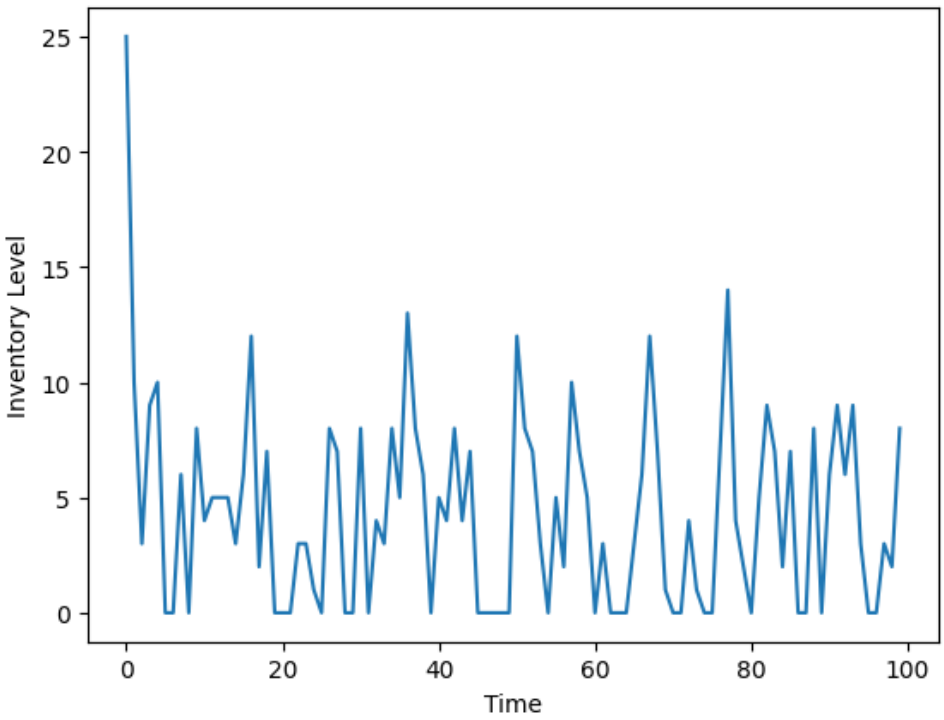

Figure 16: Inventory time series

13 Policy Results

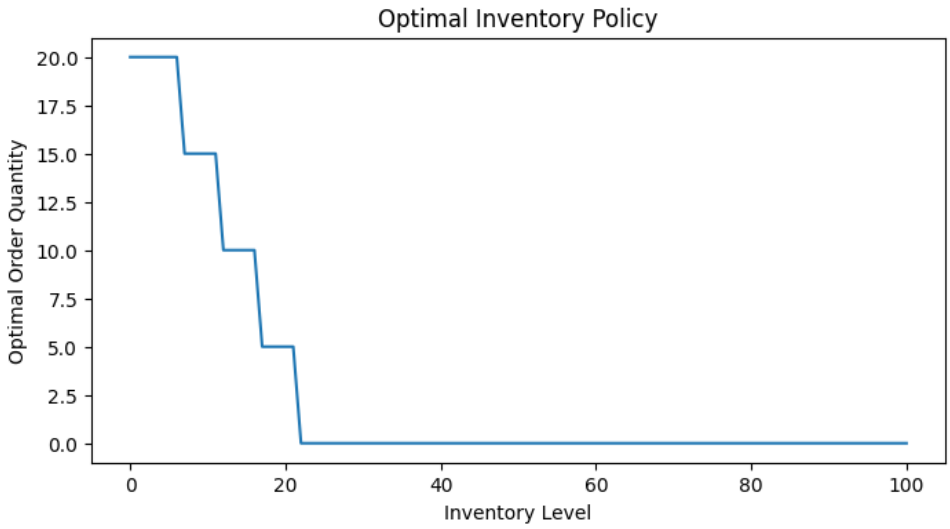

Figure 17: Optimal policy

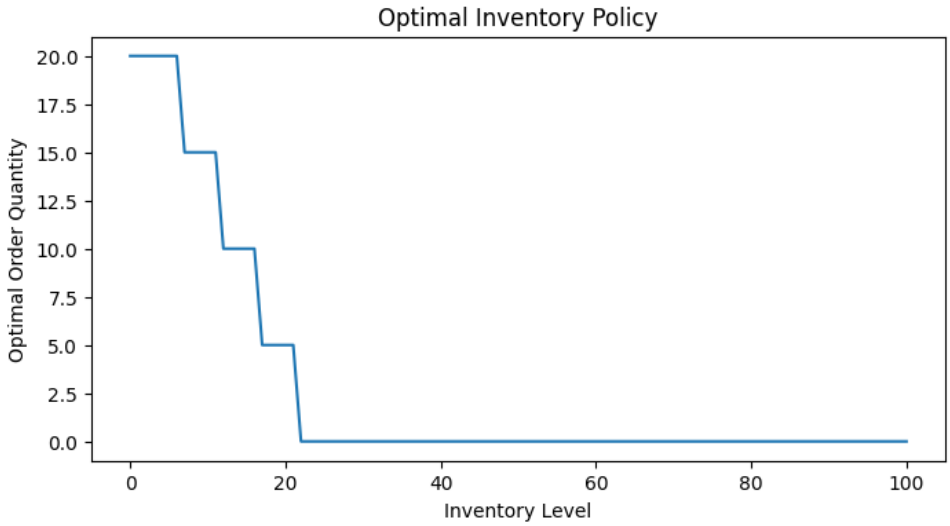

Figure 18: Optimal policy visualization

14 Markov Demand Simulation

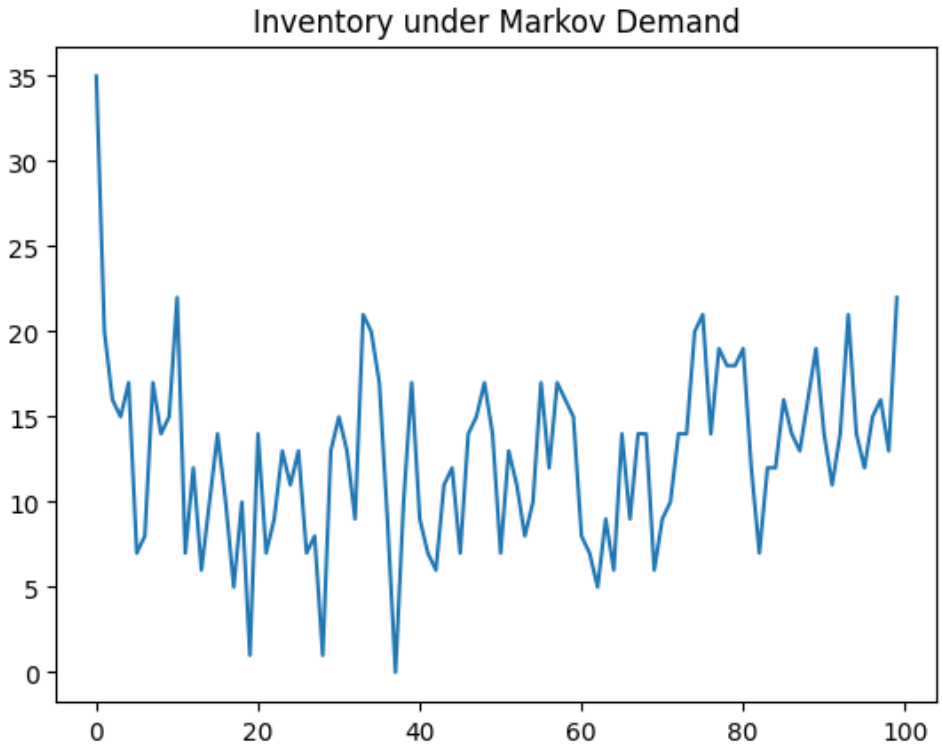

Figure 19: Inventory under Markov demand

15 Optimal Ordering

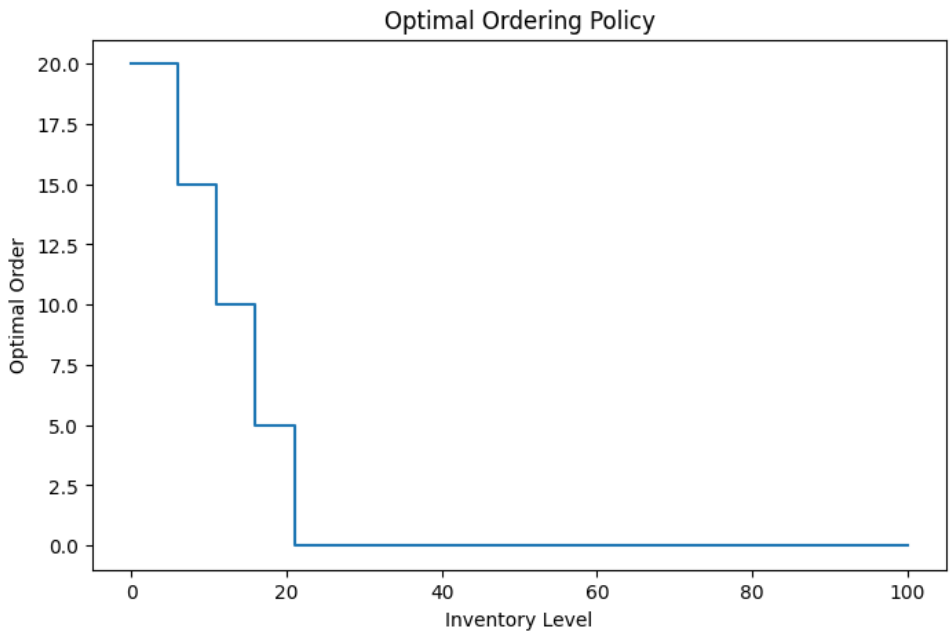

Figure 20: Optimal ordering

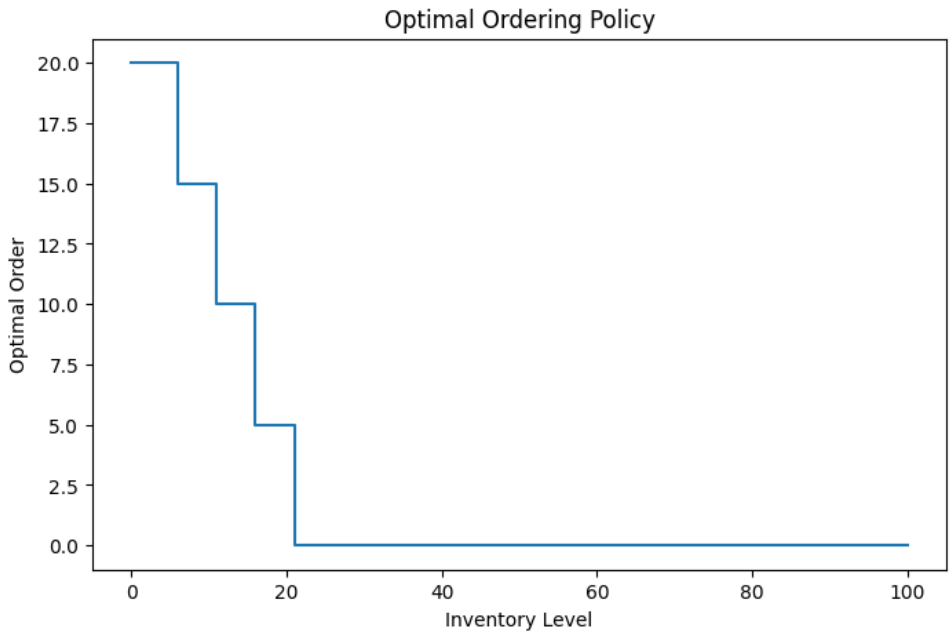

Figure 21: Optimal ordering policy

16 Service Level Analysis

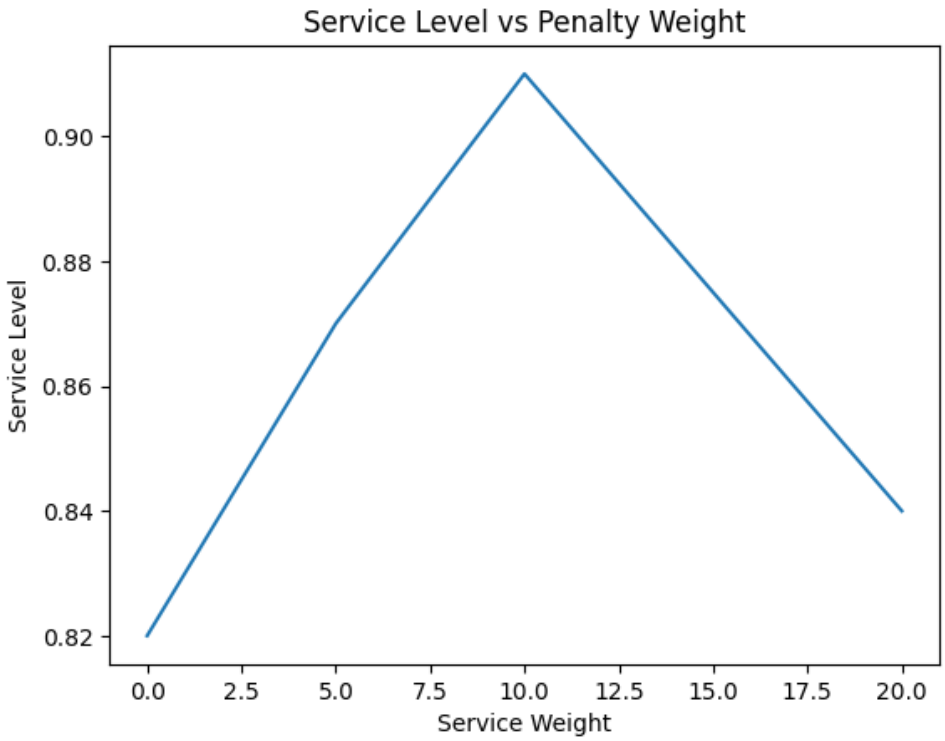

Figure 22: Service level vs penalty

17 Final Results

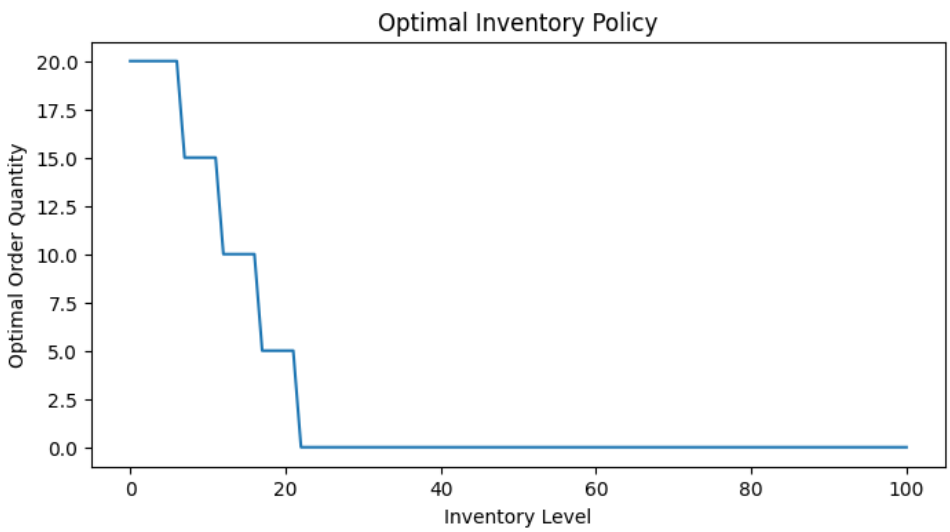

Figure 23: Final optimal policy

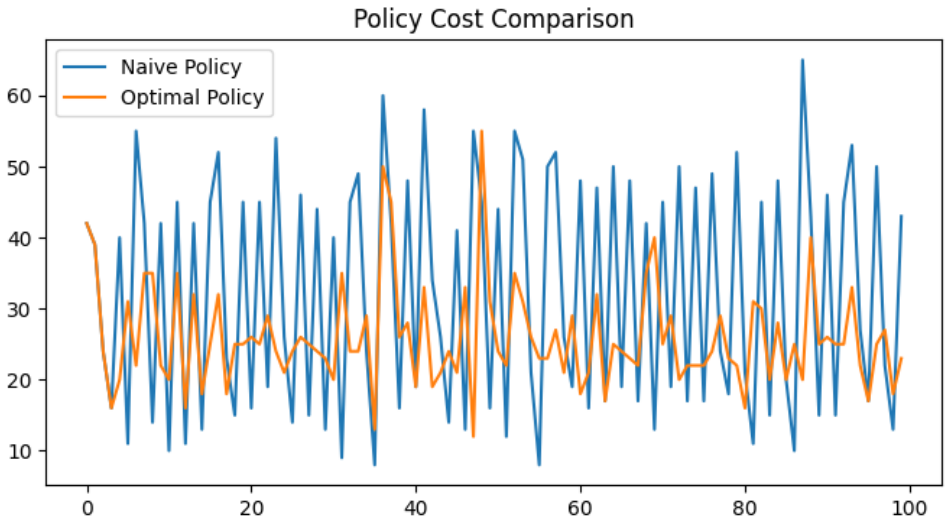

Figure 24: Policy cost comparison

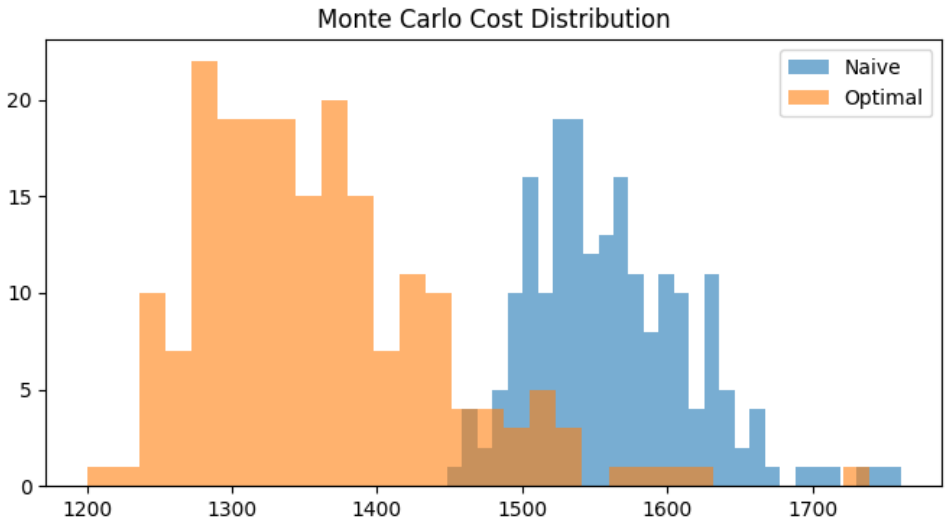

Figure 25: Monte Carlo cost distribution

18 Conclusion

This work demonstrates how stochastic optimization and Markov Decision Processes can improve inventory control decisions under uncertain demand. The optimized policy consistently outperforms naive ordering strategies and reduces long-run operational cost.

Future work could extend the model to multi-echelon supply chains, capacity constraints, and reinforcement learning based inventory policies.